



Pharm*Achieve*

ATRIAL FIBRILLATION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

4F-PCC	4-Factor Prothrombin Complex Concentrate	CYP	Cytochrome P450	NVAF	Non-Valvular Atrial Fibrillation
ASA	Acetylsalicylic Acid	CXR	Chest X-Ray	OAC	Oral Anticoagulant
AAD	Antiarrhythmic Drug	DHP	Dihydropyridine	PAD	Peripheral Artery Disease
ACS	Acute Coronary Syndrome	DOAC	Direct Oral Anticoagulant	PCI	Percutaneous Coronary Intervention
ADR	Adverse Drug Reaction	DVT	Deep Vein Thrombosis	PE	Pulmonary Embolism
AF	Atrial Fibrillation	ECG	Electrocardiogram	PGP	P-glycoprotein
ASA	Acetylsalicylic Acid	GI	Gastrointestinal	PK	Pharmacokinetic
AV	Atrioventricular	HTN	Hypertension	PO	Per Os (Oral)
AVJ	Atrioventricular Junction	HR	Heart Rate	QOL	Quality of Life
BB	Beta Blocker	INR	International Normalized Ratio	SCr	Serum Creatinine
BID	Twice Daily	K⁺	Potassium	SR	Sinus Rhythm
BP	Blood Pressure	LFT	Liver Function Test	T4	Thyroxine
BPM	Beats Per Minute	LV	Left Ventricular	TdP	Torsades De Pointes
C/I	Contraindication	LVEF	Left Ventricular Ejection Fraction	TIA	Transient Ischemic Stroke
CAD	Coronary Artery Disease	Mg²⁺	Magnesium	TSH	Thyroid-Stimulating Hormone
CCB	Calcium Channel Blocker	MOA	Mode of Action	VKORC1	Vitamin K Epoxide Reductase Complex 1
CCS	Canadian Cardiovascular Society	Non-DHP CCB	Non-Dihydropyridine Calcium Channel Blocker	VT	Ventricular Tachycardia
CHF	Congestive Heart Failure	NSAID	Nonsteroidal Anti-inflammatory Drug		
CKD	Chronic Kidney Disease				
CrCl	Creatinine Clearance				

ATRIAL FIBRILLATION

- Type of **supraventricular tachycardia**: irregularly irregular heartbeats (no repetitive patterns on ECG) + no distinct P waves on ECG
- **Atria chaotically contract** and are not coordinated with ventricles to efficiently pump blood leading to ↓ cardiac output, ↑ stroke risk
- Most common pathologic arrhythmia

CLASSIFICATIONS

Paroxysmal /self-terminating	• Episode lasting more than 30 seconds • Terminating within 7 days (spontaneously or with intervention)
Persistent	• Episode lasting longer than 7 days but less than 12 months • Sinus rhythm restored by cardioversion or medications
Long-standing persistent	• Continuous episode lasting longer than 12 months • May still be able to try and restore (and maintain) sinus rhythm
Permanent /chronic	• Decision made not to pursue sinus rhythm restoration

ATRIAL FIBRILLATION

SYMPTOMS

- Palpitations
- Tachycardia
- Dyspnea
- Light headedness
- Syncope (rare)
- Reduced exercise capacity

RISK FACTORS

- HTN
- Underlying heart disease
- Advancing age
- Sex (male >female)
- Diabetes
- Obesity
- Obstructive Sleep Apnea
- Hyperthyroidism
- Electrolyte abnormalities ($\downarrow K^+$, $\downarrow Mg^{2+}$)

DRUG CAUSES

- Alcohol
- Caffeine
- Digoxin
- Tobacco
- Antiarrhythmic drugs
- Stimulants
- Diuretics (electrolyte abnormalities)

GOALS OF THERAPY

- ✓ Relieve signs and symptoms of atrial fibrillation
- ✓ Control ventricular response rate
 - ✓ Improves symptoms
 - ✓ Does NOT restore normal sinus rhythm
 - ✓ ↓ risk adverse effects compared to restoring SR due to antiarrhythmic use
 - ✓ Rate vs rhythm control have similar risks of stroke/mortality (AFFIRM trial)
- ✓ Prevent stroke and AF recurrence
- ✓ Improve symptoms, functional capacity and quality of life
- ✓ Restore normal sinus rhythm if possible
- ✓ Minimize drug-related adverse effects

HASBLED & CHADS₂-65 SCORES

HAS-BLED SCORE:

Assess patient's bleeding risk and safety of anticoagulant use in AF

- HTN (defined as SBP above 160) (1 point)
- Abnormal liver function (1 point)
- Abnormal kidney function (1 point)
- Prior stroke or bleeding, labile INR (1 point each)
- Elderly >65 years (1 point)
- Drugs (i.e. concomitant ASA or NSAIDs) and/or alcohol (1-2 points)

0-2 = low risk of bleed
≥3 = high risk of bleed

CHADS₂-65 SCORE:

Determine stroke prevention medication to use in AF (used in the Canadian guidelines)

- CHF (1 point)
- HTN (1 point)
- Age ≥ 65 (1 point)
- Diabetes (1 point)
- Stroke/TIA (2 points)

0 = low risk
1 = moderate risk
>1 = high risk

CHADS₂ SCORE:

Determine stroke prevention medication to use in AF (original, but no longer in guidelines)

- CHF (1 point)
- HTN (1 point)
- Age ≥ 75 (1 point)
- Diabetes (1 point)
- Stroke/TIA (2 points)

CHA₂DS₂-VASc SCORE:

(used in American guidelines)

- CHF (1 point)
- HTN (1 point)
- Age > 75 (2 points)
- Diabetes (1 point)
- Stroke/TIA (2 points)
- Vascular disease (1 point)
- Age 64-74 (1 point)
- Sex category - Female (1 point)



The risk-benefit ratio almost always favours anticoagulation unless the risks for bleeding are very high and the risk of thromboembolism is very low

STROKE PREVENTION

Consider and modify factors influencing **risk of bleeds** during:

- **OAC treatment:** HTN, anti-platelets, NSAIDs, corticosteroids, excessive alcohol, labile (fluctuating) INRs
- **DOAC treatment:** low creatinine clearance, age >75, low body weight (<60 kg)

EXAMPLES OF ANTICOAGULANTS

- **DOAC:** dabigatran, rivaroxaban, apixaban, edoxaban
- **OAC:** DOAC, warfarin

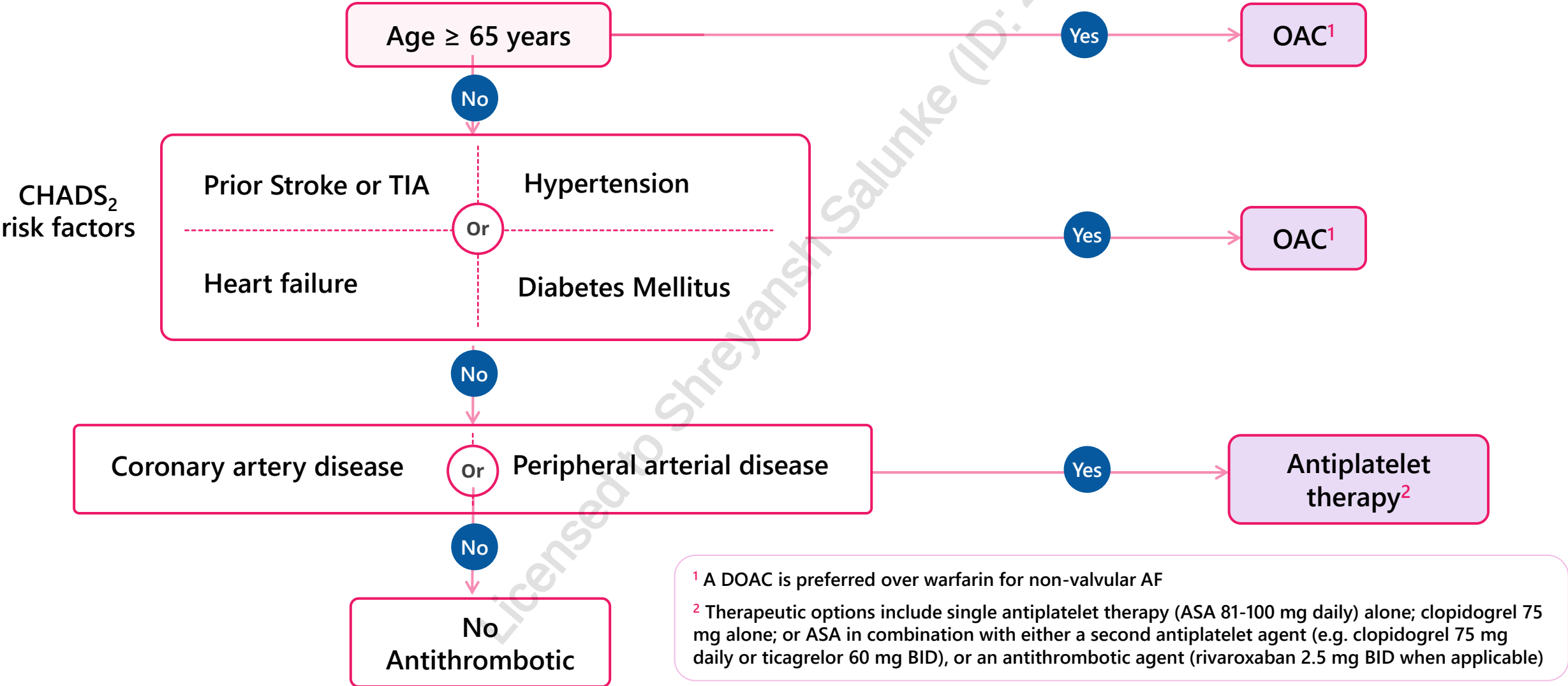
NVAF: DOACs preferred over warfarin

***Valvular (Ex: Heart valve replaced) AF:** warfarin only

- Associated with moderate to severe mitral stenosis and/or presence of **MECHANICAL** heart valve

CCS ALGORITHM (CHADS-65)

OAC Therapy in AF



PHARMACOLOGICAL MEASURES - STROKE PREVENTION

Drug	Dose [memorize!]	PK Parameters	Renal Adjustment	Reversal Agent
Dabigatran	150 mg PO BID 110 mg PO BID in all patients ≥ 80, or if age ≥ 75 and other bleeding risk factors including CrCl 30-50 mL/min	PGP substrate	- CrCl > 30: 150 mg PO BID or 110 mg PO BID - CrCl < 30: Avoid	Idarucizumab
Rivaroxaban	20 mg PO daily with food 15 mg PO daily if CrCl 30-50 mL/min	CYP 3A4 and PGP substrate	- CrCl 30-50: 15 mg PO daily (measure CrCl every 6 months) - CrCl 15-29: Use 15 mg PO daily with caution (if used, measure CrCl every 3 months) - CrCl < 15: Avoid	Andexanet alfa
Apixaban	5 mg PO BID 2.5 mg PO BID if 2 of following: age ≥ 80, weight ≤ 60 kg, SCr ≥ 133 μmol/L	CYP 3A4 and PGP substrate	- CrCl 15-29 mL/min: Use with caution - CrCl < 15 mL/min or undergoing dialysis: Avoid use	Andexanet alfa
Edoxaban	60 mg PO daily 30 mg PO daily if CrCl 15-50 mL/min or ≤ 60 kg or Pgp inhibitor	CYP 3A4 and PGP substrate	- Do not use if CrCl > 95 mL/min - CrCl 15-50: 30 mg PO daily - CrCl < 15: Avoid	Andexanet alfa (Off-label use) May use 4F-PCC* instead

* 4F-PCC (Octaplex® or Beriplex®) is not a specific reversal agent but may aid in hemostasis. Can be considered as an alternative if reversal agents are not available/indicated



Weigh benefit against risk of hemorrhage!

WARFARIN

- MOA: **inhibits VKORC1** → decreases functional vitamin K reserves → reduces synthesis of active clotting factors
 - Affects factors 2,7,9,10 and protein C and S
- **1st line for valvular AF** → DOACs have not shown benefit in this population
- **1st line for moderate to severe mitral stenosis**
- ONSET: Can take **24-72 hours** to see the initial anticoagulant effect
- HALF LIFE: 20-60 hours
- METABOLISM: Primarily a CYP2C9 substrate but also a substrate of CYP 2C8, 2C19, 1A2, 3A4
 - **Large amount of drug interactions!**
 - Heavy alcohol use can reduce serum warfarin concentrations.
 - **Drug interactions aren't always contraindications** → monitor INR and adjust dose accordingly
- Monitored by INR → **target for AF is 2.0-3.0**
- Dose is personalized to every individual → use a dosing nomogram upon initiation

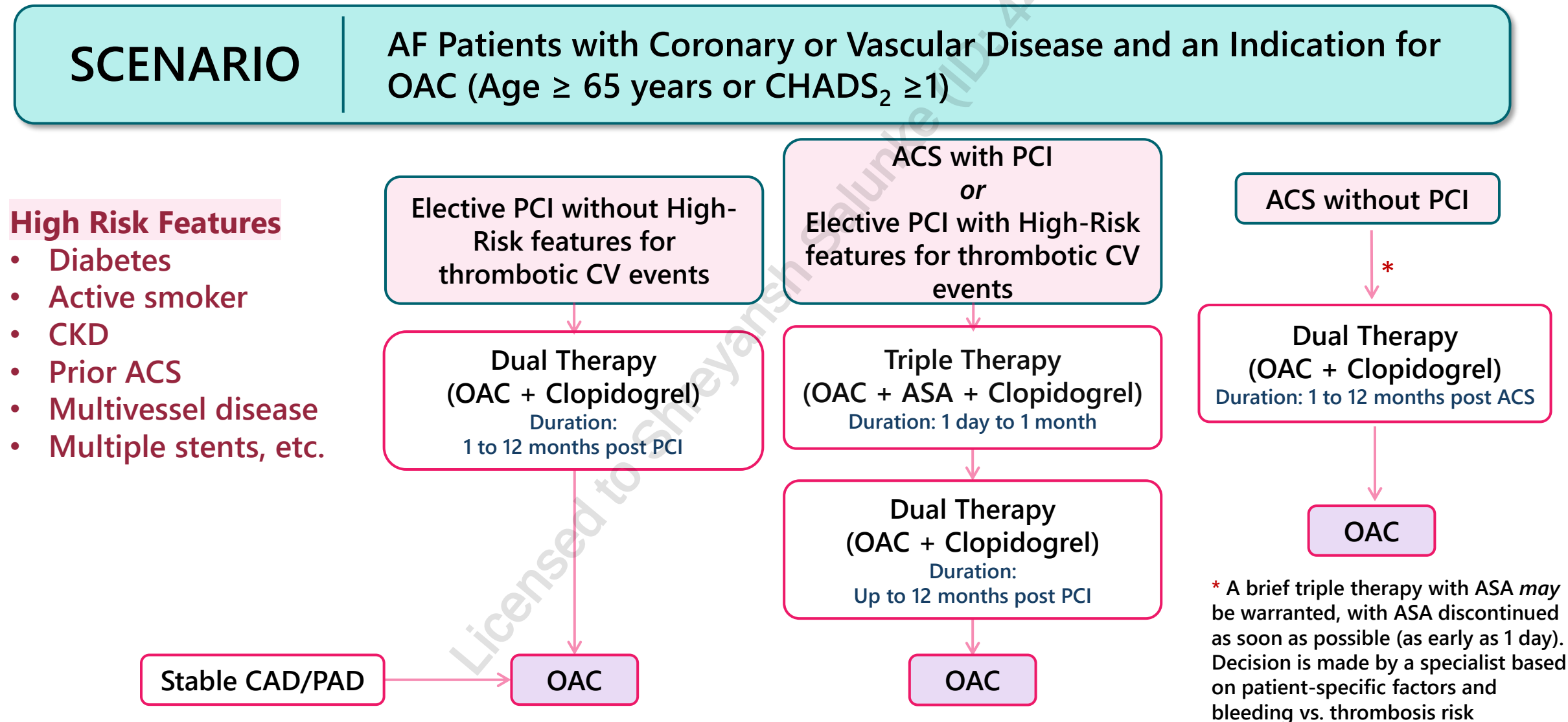
ANTIPLATELET AND ANTICOAGULANT THERAPY

ANTIPLATELETS	ANTICOAGULANTS
MOA: Prevent platelets from clumping to form clots Commonly used in CAD, PAD, secondary prevention of stroke, prevention of stent thrombosis in PCI	MOA: Prevent fibrin from forming (works on coagulation cascade) Commonly used in DVT, PE, AF

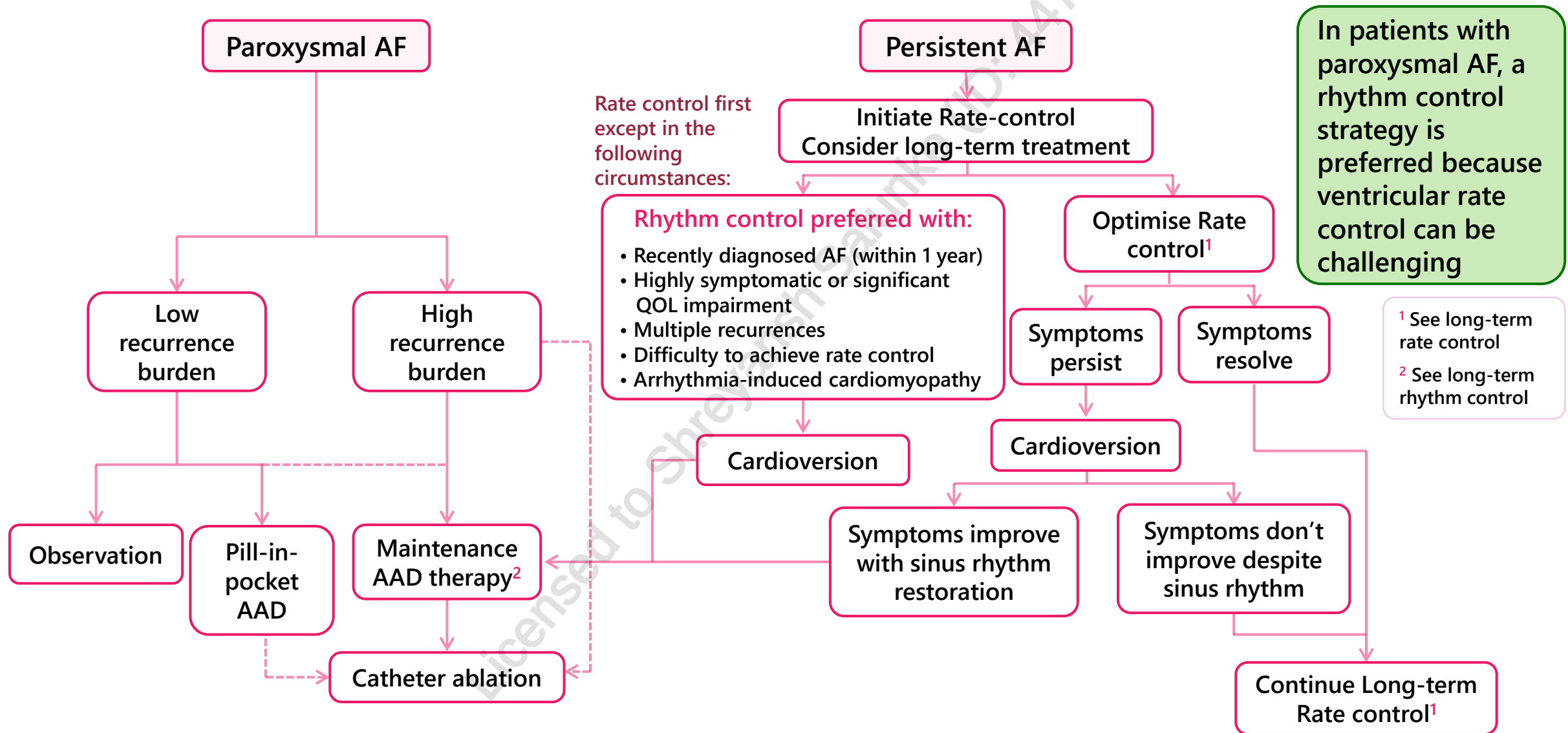
What happens when a patient has an indication for antiplatelets AND anticoagulants?

- For patients with concomitant AF and CAD/PAD, must balance bleeding versus thrombotic risk factors
- Up to 20-30% of AF patients have CAD (a significant portion also requiring PCI)
 - DOAC required for AF stroke prevention in addition to antiplatelet for prevention of coronary events and stent thrombosis
- Antiplatelets and anticoagulants can be used together IF INDICATED
 - Tips: Consider PPI use, avoid prasugrel and ticagrelor with OACs, use a lower target INR (e.g. 2.0-2.5) with warfarin, avoid NSAIDs or other drugs with a bleeding risk, strict BP control, delay non-urgent procedures until dual or triple therapy no longer required

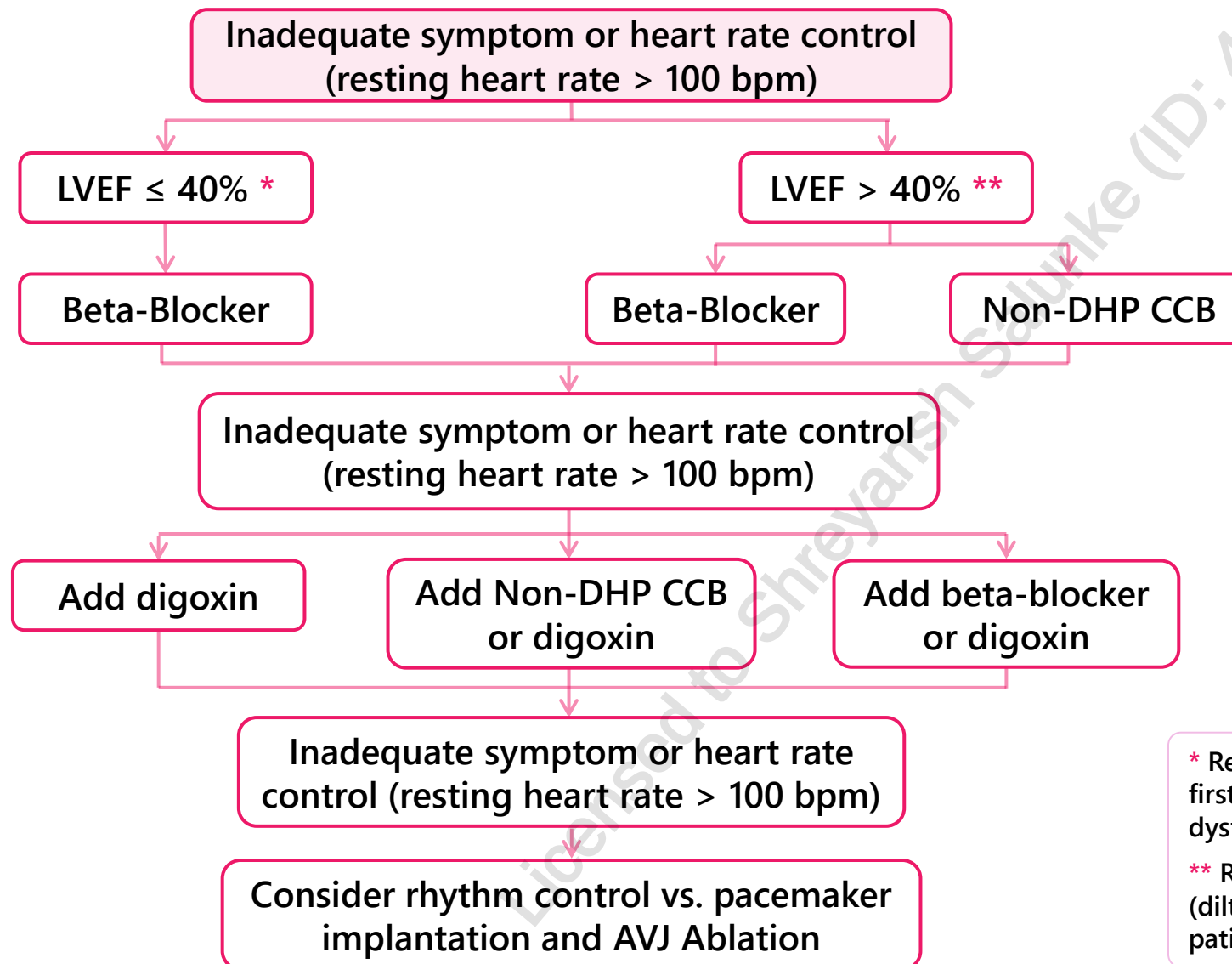
TREATMENT ALGORITHM



APPROACH TO RATE AND RHYTHM MANAGEMENT OF AF



LONG-TERM RATE CONTROL



Target resting heart rate
< 100 bpm adequate if
symptoms resolved. No
good evidence for
stricter rate control!!

* Recommend using bisoprolol, carvedilol or metoprolol first-line for rate control in patients with significant LV dysfunction (LVEF less than or equal to 40%)

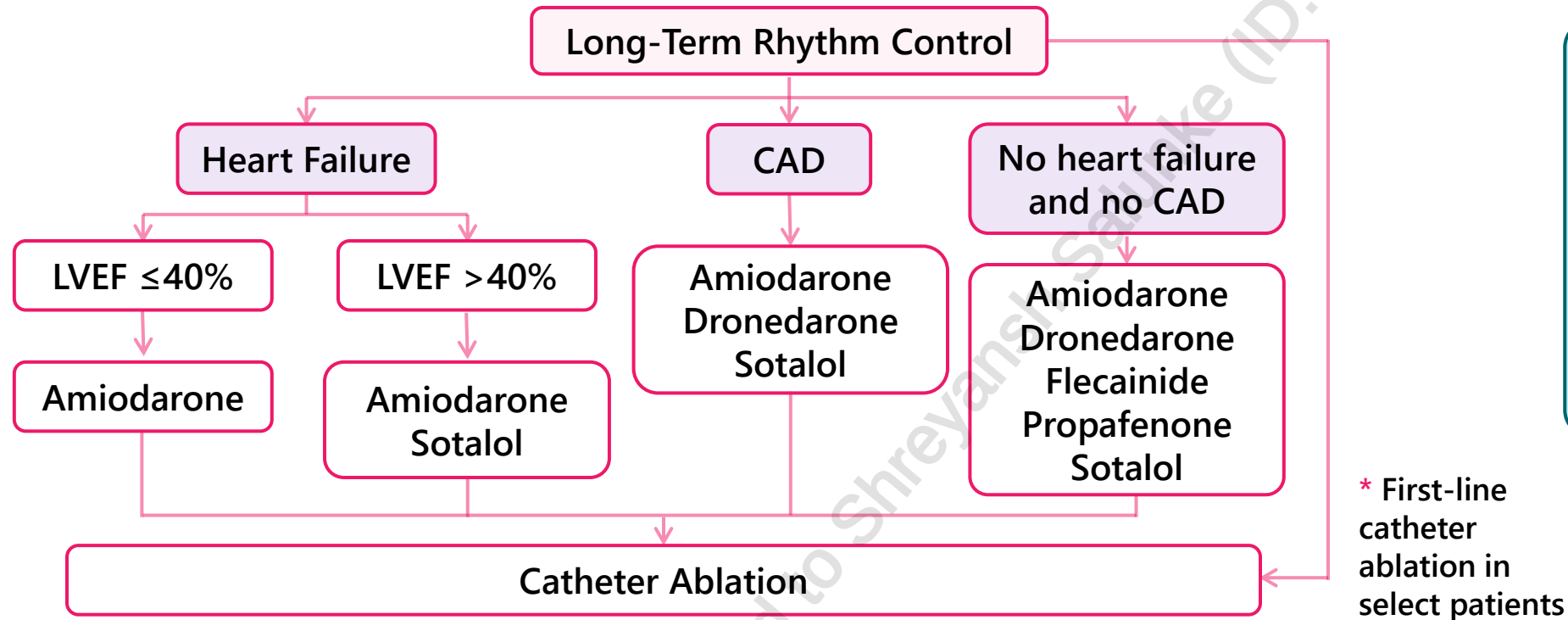
** Recommend using beta-blockers or non-DHP CCBs (diltiazem or verapamil) first-line for rate control in patients without significant LV dysfunction (LVEF > 40%)

PHARMACOLOGICAL MEASURES - RATE CONTROL

Drug	Mechanism	Toxicity	Comments
Beta Blockers	Bind to beta 1 and 2 receptors to inhibit action of epinephrine and norepinephrine	• Bradycardia • Hypotension • Dyspnea • Fatigue	• Monitor carefully in patients with diabetes; use with caution in patients with HF or lung disease (e.g. asthma) • Use β_1 selective BB (cardio selective)
Non-DHP Calcium Channel Blockers	Block L-type Ca channels in cardiac muscle and nodal tissue to slow down HR and contractility by reducing Ca influx	• Bradycardia • Hypotension • Constipation with verapamil	• Additive negative chronotropic effects with amiodarone, beta-blockers, digoxin
Digoxin	Increases force of myocardial contraction (positive inotrope) but also increases vagal tone to slow HR by decreasing conduction through AV node	• Bradycardia • Nausea/Vomiting • Visual disturbances • Proarrhythmic	• Reserved for patients with symptomatic HF and low ejection fraction. • Check serum potassium, correct hypokalemia if present • With beta-blockers, calcium channel blockers, amiodarone, dronedarone, propafenone, quinidine: reduce digoxin dose by 25–50%.

RHYTHM CONTROL

Maintenance AAD therapy



Maintain sinus rhythm:

- Amiodarone
- Dronedarone
- Flecainide
- Propafenone
- Sotalol



- Class Ic agents e.g. flecainide, propafenone should be avoided in CAD and should be combined with AV-nodal blocking agents → increased risk of atrial flutter
- Sotalol should be used with caution in those at risk for torsades de pointes (e.g. female, age > 65 years, concomitant potassium-wasting diuretics) and in LV hypertrophy

PHARMACOLOGICAL MEASURES - RHYTHM CONTROL

Drug	Mechanism	Toxicity	Comments
Flecainide	Class IC	- Ventricular tachycardia - Bradycardia - rapid ventricular response to AF or atrial flutter - Blurred vision - Tremor	Contraindicated in CAD or LV dysfunction - Should combine with AV nodal blocking agent - Metabolized by CYP 2D6
Propafenone	Class IC	- Ventricular tachycardia, - Bradycardia - rapid ventricular response to AF or atrial flutter - dysgeusia	Contraindicated in CAD or LV dysfunction - Should combine with AV nodal blocking agent - Metabolized by CYP 2D6 and CYP 3A4. - Reduce digoxin dose by 25-50%
Amiodarone	Multichannel blocker	- Photosensitivity - Bradycardia - GI symptoms, - Thyroid dysfunction, hepatic toxicity, neuropathy, tremor, pulmonary toxicity, TdP	- Has a long half life, Most adverse effects are dose and duration-related - Interacts with diltiazem, quinidine, procainamide, among others - Measure TSH at baseline and then every 3-6 months - Monitor Transaminases every 6 months - Chest Xray annually

PHARMACOLOGICAL MEASURES - RHYTHM CONTROL

Drug	Mechanism	Toxicity	Comments
Dronedarone	Class III	• GI symptoms • Bradycardia • Hepatic toxicity • Pulmonary fibrosis	• Do not use for rate or rhythm control in patients with history of CHF or LVEF <40% • Caution with concomitant digoxin use • Monitor LFTs • Similar in structure to amiodarone, but different adverse effect profile • Inhibits CYP 3A4
Sotalol	Class III but also a beta blocker at lower doses	• Hypotension • TdP and QTc prolongation • Bradycardia	• Avoid in patients at high risk of TdP VT especially females >65 years who are also taking diuretics or have renal insufficiency • Perform baseline ECG and 48-72 hours after outpatient initiation to monitor for QT prolongation • Contraindicated in patients with LVEF ≤40%

"PILL-IN-THE-POCKET" ANTIARRHYTHMIC DRUG THERAPY

Indications	• Symptomatic patients with sustained AF episodes (e.g. >2 hours) that occur less frequently than monthly • Absence of severe or disabling symptoms during an AF episode (e.g., fainting, severe chest pain, or breathlessness) • Ability to comply with instructions and proper medication use
C/I	• Significant structural heart disease (e.g. LVEF <50%, active ischemic heart disease, severe left ventricular hypertrophy) • Abnormal conduction parameters (e.g., QRS duration >120 msec, PR interval >200 msec) • Sinus node dysfunction or advanced AV block • Hypotension (systolic BP <100 mmHg)
How to take and referral	• Immediate release PO AV nodal blocker (either diltiazem, verapamil, or metoprolol) should be administered 30 minutes prior to the administration of a class 1c antiarrhythmic (e.g. flecainide or propafenone) • Following antiarrhythmic drug administration, patients should rest in a supine or seated position for the next 4 hours, or until the episode resolves • Patients should present to the emergency department in the event that: - The AF episode did not terminate within 6-8 hours - They felt unwell after taking the medication at home (e.g. subjective worsening of the arrhythmia following antiarrhythmic drug ingestion) - More than one episode occurred in a 24-hour period (patients are advised not to take a second "pill-in-the-pocket" dose) - If the AF episode was associated with severe symptoms (e.g. significant dyspnea, chest pain, pre-syncope, or symptoms of stroke), even in the absence of "pill-in-the-pocket" use

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
HR	Resting < 100 bpm	Days
ECG if rhythm control	Normal SR	Days

Safety Endpoints		
Parameter	Degree of Change	Timeframe
ADR from anti-arrhythmic / rate control / anticoagulation (ex. bradycardia, bleeding)	Prevent	Duration of therapy
Stroke	Prevent	
Creatinine, Platelets, hemoglobin (anticoagulants), LFTs, TSH/T4, CXR, eye exam, ECG (amiodarone)	Within normal range	
INR (if on warfarin)	2.0-3.0	

CASE 1

AF presents to the ED with palpitations and some dizziness. AF is a 78-year-old female with an 8-day history of palpitations, worsening for the past 24 hours with decreased exercise ability. An ECG in the ED showed atrial fibrillation. AF has a past medical history of HTN and type 2 diabetes mellitus for which she takes ramipril 5 mg PO daily, metformin 1000 mg PO BID and gliclazide MR 30 mg PO daily, respectively. She smokes 10 cigarettes per day, drinks 5 alcohol drinks on weekends, and has 2-3 cups of coffee per day. She has a penicillin allergy from when she was a child that presented as a rash.

Objective findings:

- BP=118/75 mmHg, HR=133 bpm
- SCr=65 $\mu\text{mol/L}$ (reference: 44-97 $\mu\text{mol/L}$), CrCl 104 mL/min
- TSH=1.1 mIU/L (reference: 0.5-4 mIU/L)
- K^+ = 3.6 mmol/L (reference: 3.6-5.0 mmol/L)
- Weight= 75 kg

1. What is AF's CHADS₂-65 score?

- a) 1
- b) 2
- c) 3
- d) 4



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2. Which of the following treatment strategies is the most appropriate?

- a) Initiate rate control with metoprolol
- b) Initiate rhythm control with amiodarone
- c) Initiate rate control with digoxin
- d) Catheter ablation



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3. Which of the following are monitoring parameters of Amiodarone?
- a) Eye Exams
 - b) LFTs
 - c) TSH
 - d) All of the above



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CHANGE LOG

July 2025

- Formatting changes made throughout; legend and references updated
- Slide 4: Added “or with intervention” to paroxysmal AF definition
- Slide 5: Minor rephrasing changes, added ‘hyperthyroidism’ to risk factors
- Slide 10: Harmonized doses and dose adjustments to CCS guidelines
- Slide 12: Added Tips on OAC use with antiplatelets
- Slide 16: Added “potassium-wasting” to diuretics as a risk factor for TdP, added “hypertrophy” to sotalol footnote, changed “Class 1” to “Class 1c” in footnote
- Slide 19: Changed sotalol “Caution” to C/I for LVEF $\leq 40\%$, updated monitoring
- Slides 21-22: Added details to case

September 2025

- Formatting changes made throughout, reorganization of material and slide order
- Slide 12: Added information on mechanism of action
- Slide 16: New
- Slides 18 and 19: Added more information to drugs
- Slide 24: Added new question 3 to case 1
- Slide 25: Updated references

January/February 2026

- Slide 2: Added new NAPRA competencies
- Slide 5: Added sleep apnea and obesity to risk factors
- Slide 10: Added off-label use for Andexanet alfa with Edoxaban and added note on 4F-PCC
- Slide 13: Added note regarding algorithm for ACS without revascularization
- Slide 25: Updated references